

Jijo Joy

ROI Full Driver's License | SafePass(SOLAS learning work)

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I Mechanical engineer who possesses over three years of work experience in production planning, repairs and over five years of educational experience in mechanical and aerospace discipline. I have multi-disciplinary design knowledge and experience in mechanical design, diagnostic as well as performance enhancement with special reference to AutoCAD Drawing tool creating new systems and improving existing systems. I have also fixed and maintained out of board engines and handled mechanical and aircraft components at AL JNZ Trading, Jet Airways and Air India Engineering Services. I have completed my professional education in Aeronautical Engineering and after that, I have done a master's degree in Big Data Analytics thus, I have balanced and competent skills to address engineering issues.

Core Skills

- AutoCAD
- Thermodynamics
- Fluid Mechanics
- Autodesk CFD Software
- Aviation ERP – RAMCO, AMOS
- Maintenance Skills – SI Engine (2 stroke and 4 stroke piston engine), CI Engine (2 stroke and 4 stroke diesel engine), Turbofan Engine
- Working Principles – Reciprocating Engines, Rotary Engines, Turbofan Engine, Turboshift Engine, Turboprop Engine, Ramjet Engine, Scramjet Engine.
- Mechanical Systems – Drivetrain System, Exhaust System, Steering System, Suspension System, Transmission System, Cooling System, Pneumatic System, Bleed Air System, Hydraulic Systems, Turbo System, Supercharger System, Fuel Systems (Rocket, Aircraft, Vehicles), Auto Pilot System, Avionics Systems, Landing Gear Systems, Braking Systems, Auxiliary Power Systems
- Air Regulatory Authority – DGCA, EASA
- SHEQ, LOTO
- Python for trend analysis and optimization.
- Microsoft365 - MS Word, MS Excel, Outlook, Teams, Sharepoint
- Presentation Software - Tableau, MS PowerPoint
- Communication skills for coordinating with clients, contractors, and regulatory authorities.

Work Experience

Panda Power | Dublin, Ireland

MECHANICAL ENGINEER | CHP Engines & CH₄ Gas Pipes | Dec 2024 to June 2025

- Conducted CH₄ gas pipe balancing to identify and isolate oxygen-leaking pipelines, ensuring safe and efficient biogas distribution.
- Performed maintenance on CHP engines, working with components including heat exchangers, heat pumps, hydraulic and pneumatic systems.
- Carried out scheduled inspections: E10 (50hrs), E15 (oil change), E30 (1500hrs), E40 (3000hrs) on Deutz TBG 616, MWM TCG 3016 V8, TCG 2020 V12, TCG 2016 V16, and CAT CG132 engines.
- Executed oil changes and inspections at 1000hr, 2000hr, 4000hr, and 10,000hr intervals on Jenbacher JGC 420 GS-L.L and Jenbacher 312/320 engines.
- Diagnosed and resolved mechanical faults in real-time, minimizing downtime and improving overall system reliability.
- Operated CHP engines using biogas produced from county council facilities to generate electricity, synchronized output, and exported it to the ESB mains (national electricity grid).
- Followed SHEQ procedures and Lockout/Tagout (LOTO) protocols to maintain a safe working environment and ensure compliance with health and safety standards.

AL JNZ TRADING AND SERVICES | Sur, Oman

MECHANICAL ENGINEER | Repair/Production Planning Department(PPD) | June 2019 to June 2021

- Interpreting diagnostic information, conducting functional tests, and pinpointing the root cause of issues for outboard engines.
- Performed visual inspections, assessing wear and tear, leaks, corrosion, and other visible signs of problems.
- Oversee disassembling, inspecting, replacing worn parts, and reassembling engines according to manufacturer 2D CAD specifications.
- Fixing and fine-tuning carburettors/fuel injectors, fuel pumps, starter motors, water pumps, and other key components for optimal performance.
- Utilizing CAD to design tools and fixtures that aid in specific repair tasks, improving efficiency and precision.
- Utilizing CAD to ensure custom parts are feasible to produce according to workshop capabilities and available materials.
- Using CAD software to design and visualize custom components like mounts, brackets.

JET AIRWAYS | Mumbai, India

JUNIOR TECHNICAL ENGINEER, Engineering Repairs Department | March 2019 to June 2019

FLEET:

- o BOEING 777-300ER, 737MAX8, 737-800, 737-900
- o AIRBUS A330-200, ATR 72-600
- Tracking the fleet's hours and cycles in the AMOS system and producing MIS reports, incident/accident clearance statements, and aircraft status reports for different lessors.
- Re-delivery: We have helped Lessors receive their aircraft back after early and end-of-lease terminations.
- Have helped with several lease terminations and asset title transfers within the Jet fleet.
- Oversaw the declarations and requirements for aircraft documents for aircraft that lessees had deregistered.
- Technical specifications for the ATR, B737, B330, and B777 have been created for marketing purposes as well as possible sale and lease back.

AIR INDIA ENGINEERING SERVICES LIMITED | Mumbai, India

ON JOB TRAINEE | Repair/PPD | June 2018 to Feb 2019

FLEET:

- o BOEING 747-400, 777-200, 777-300, 787
- Have a brief and good knowledge of schedule arising plan and monthly aircraft accessories due report.
- Production Statistics-Month wise Analysis.
- Assisting in Preparation of Daily Production Schedule (DPS) for various Section of AOD shops (Electrical/Fuel/Hydraulic/Pneumatic) as per Aircraft/JOD/other Requirements
- Preparation of prioritizing list of Engines & APU components for production from concerned overhaul shops.
- Know the concept of Preparation and Revising of Hanger Visit Plan (HVP) (Long Term HVP, Short Term HVP)
- Assisting in Preparation of Daily Production Schedule (DPS) for various Section of COD shops (Wheels, brakes, oxygen bottle, nacelle & thrust reverser) as per Aircraft/JOD/other requirements.
- Using Boeing performance software (BPS/APM), analyse and track the engine's cruise performance for the following aircraft:
Boeing B747-400 Aircraft fitted with PW 4056 Engines.
Boeing B777-300ER Aircraft fitted with GE90-115 Engines.
Boeing B777-200LR Aircraft fitted with GE90-110 Engines.
Boeing B737-800 NG Aircraft fitted with CFM 56-7B Engines.
- Regularly monitor the fuel factor of the above entire Boeing fleet in Air India and Air India Express.
- Help with the preparation of the Boeing aircraft engine performance report for CAMO.
- Keep up the engineering static reports and fleet performance reports.

- Knows how to work on latest version of RAMCO aviation software V.5.8.

HINDUSTAN AERONAUTICAL LIMITED | Bangalore, India

Industrial Training | Helicopter Division | June 2015

FLEET:

- o Aerospatiale SA 316B Alouette III — HAL Chetak, Chetan Variants
- o Dhruv — Advanced light helicopter
- o Rudra — (in production) Armed and reconnaissance version of Dhruv
- o LCH (Light Combat Helicopter)

Languages English, Hindi, Malayalam, Tamil, Arabic

Education

Hindustan University, Chennai India B. Tech Aeronautical Engineering - Nov2017

Griffith College, Dublin MSc in Big Data Analytics and Management - Nov2023

Griffith College, Dublin Higher Diploma in Computing Science - Sep2022